

CHANDRASHEKAR BALA

Cybersecurity Researcher | Offensive Security | VAPT & Security Engineering

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SUMMARY

Cybersecurity professional focused on security assessment, vulnerability research, and security engineering across applications, networks, systems, and emerging technologies. Hands-on work spans penetration testing, VAPT, web application security, network and wireless security, vulnerability validation, exploit research, privilege escalation, post-exploitation analysis, malware analysis, digital forensics, threat intelligence, and security operations. I approach security from attacker and defender perspectives—reconstructing attack paths, validating what is actually exploitable, analyzing evidence and impact, and translating findings into remediation, detection, and engineering improvements. Currently a Cybersecurity Intern at NETWORKWALKS Technologies while continuing independent security research and systems-level engineering.

EXPERIENCE

Cybersecurity Intern — NETWORKWALKS Technologies

Sep'26 – Present

VAPT | Network Security | Security Operations | Incident Response | Digital Forensics

- Work across network security, vulnerability assessment, penetration testing, security operations, incident response, digital forensics, and risk assessment activities.
- Perform reconnaissance, vulnerability validation, technical investigation, evidence review, and security documentation across assigned cybersecurity work.
- Apply offensive-security and investigative techniques to identify weaknesses, reconstruct attack paths, analyze security events, assess impact, and document remediation considerations.

Cybersecurity Researcher — Independent

Jul'24 – Present

Independent Cybersecurity Research & Security Engineering

- Conduct hands-on research across offensive security, vulnerability assessment, web application security, network and wireless security, threat intelligence, malware analysis, digital forensics, and security engineering.
- Perform reconnaissance, service enumeration, vulnerability assessment, controlled exploitation, privilege escalation, post-exploitation analysis, and attack-path reconstruction across isolated security environments.
- Test web applications using Burp Suite and OWASP ZAP across SQL injection, XSS, authentication, authorization, access control, session behavior, and HTTP request/response workflows.
- Analyze OSINT, domains, IP addresses, infrastructure, threat reports, indicators, and attacker TTPs; use MITRE ATT&CK to structure adversary behavior analysis.
- Conduct network traffic and forensic analysis using Wireshark, Ghidra, Volatility, GDB, and Windows artifacts; investigate suspicious files, execution behavior, filesystem activity, and disk evidence.
- Develop Python and Bash tooling for reconnaissance, repeatable security analysis, proof-of-concept development, and security research workflows.

FEATURED SECURITY WORK

Mediroza General Hospital — Authorized Black-Box Web Application Security Assessment Sep'26 – Oct'26

Web Application Security | Burp Suite | Nmap | Manual Validation | CVSS/CWE

- Conducted an authorized black-box assessment of the in-scope web application and reconstructed the attack path from reconnaissance through demonstrated SQL injection, authentication bypass, protected report retrieval, and impact analysis.
- Documented eight security findings spanning directory exposure, internal database-backup exposure, SQL injection, verbose database errors, sensitive report retrieval, predictable storage, error-log disclosure, and technology fingerprinting.
- Produced evidence-driven findings covering exploitation conditions, severity, CWE/CVSS context, root cause, security impact, remediation, and retest considerations.

RTL8812BU / RTL8822BU Linux Driver Modernization & Wireless Security Engineering Jul'26 – Aug'26

C | Linux Kernel | GCC | Git | Kali Linux | Real Hardware

- Analyzed kernel-facing C code and implemented targeted compatibility changes across 13 source files for newer Linux kernel APIs and build requirements.
- Diagnosed compiler, kernel API, and module-build issues using Linux kernel headers, GCC, conditional compilation, Git, and kernel build workflows.
- Built and validated the modified kernel module on real RTL8812BU hardware and exercised monitor mode, packet injection, packet capture, Wireshark, and Aircrack-ng workflows.

Network Reconnaissance & Attack-Surface Analysis

Security Research

Nmap/NSE | WHOIS | DNS | TLS | HTTP | OSINT

- Performed structured reconnaissance and service validation using WHOIS, DNS tooling, certificate discovery, web fingerprinting, Nmap/NSE, TLS inspection, and evidence-preserving scan artifacts.
- Correlated host, service, web, and infrastructure data to distinguish observed exposure from hypotheses and support subsequent security assessment.

Web Application Security Research

Feb'26 – Present

PortSwigger Web Security Academy | Burp Suite | OWASP ZAP

- Investigated SQL injection, XSS, authentication, authorization, access-control, session, and HTTP behavior; reproduced findings and documented technical evidence, impact, and remediation considerations.

PROJECTS & TECHNICAL RESEARCH

Vulnerability Assessment & Privilege Escalation

Dec'25 – Present

VulnHub / Nmap / Linux / Metasploit

- Perform reconnaissance, service enumeration, vulnerability assessment, controlled exploitation, Linux privilege escalation, and post-exploitation analysis across isolated virtual environments.
- Trace attack paths from initial access through elevated privileges and document exploitation conditions, evidence, security impact, and hardening considerations.

Buffer Overflow Exploit Development

Completed

C / Linux / GDB / Memory Analysis

- Analyze stack-based buffer-overflow vulnerabilities using GDB, stack layout, registers, memory analysis, control-flow investigation, and controlled proof-of-concept workflows.

Enterprise Network Penetration Testing

Completed

Nmap / Metasploit / Linux / Network Security

- Conduct network reconnaissance, service enumeration, vulnerability assessment, controlled validation, and attack-path analysis against isolated enterprise-style environments.
- Investigate exposed services and security weaknesses, validate exploitation paths, and document technical observations and remediation considerations.

Malware Analysis & Windows Forensics Research

Security Research

Static/Dynamic Analysis / Ghidra / GDB / Volatility / Wireshark

- Conduct controlled malware-analysis and forensic research involving suspicious files, registry artifacts, filesystem activity, disk evidence, execution behavior, indicator extraction, and timeline-oriented investigation.
- Use static and dynamic analysis to examine binary behavior and correlate host and network evidence during investigation workflows.

Security Automation & Proof-of-Concept Tooling

Independent Research

Python / Bash / Linux / Security Research

- Develop Python and Bash tooling for reconnaissance, repeatable analysis, security testing, proof-of-concept development, and research automation.

TECHNICAL EXPERTISE

Offensive Security	Reconnaissance, service enumeration, vulnerability assessment, penetration testing, vulnerability validation, controlled exploitation, privilege escalation, post-exploitation, attack-path analysis
Web & API Security	OWASP Top 10, SQL injection, XSS, authentication, authorization, access control, HTTP/HTTPS, request/response analysis, session behavior, Burp Suite, OWASP ZAP
Network & Wireless	TCP/IP, DNS, DHCP, routing, switching, subnetting, CIDR, network reconnaissance, packet capture, traffic analysis, Nmap, Wireshark, Aircrack-ng, monitor mode, packet injection
Threat Intelligence	OSINT, threat intelligence, threat-report analysis, infrastructure analysis, domains/IPs, indicators, TTP analysis, adversary research, MITRE ATT&CK
Malware & Forensics	Malware analysis, static/dynamic analysis, Windows forensics, registry analysis, filesystem and disk analysis, suspicious-file investigation, memory analysis, Ghidra, Volatility
Exploit Research	Buffer-overflow analysis, C, Linux, GDB, stack and memory analysis, control-flow investigation, proof-of-concept development
Security Engineering	Linux security, kernel-facing C, Linux kernel compatibility, wireless driver engineering, GCC, Git, Python/Bash automation, technical troubleshooting
Security Operations	Security monitoring, log analysis, alert triage, IOC analysis, threat detection, suspicious-activity investigation, incident response, evidence correlation

TOOLS & TECHNICAL STACK

Application Security	Burp Suite, OWASP ZAP, HTTP/HTTPS analysis, authentication and access-control testing
Network / Discovery	Nmap/NSE, Wireshark, packet capture, service enumeration, DNS tooling, network reconnaissance, protocol analysis
Exploitation / Research	Metasploit, GDB, controlled exploitation, privilege escalation, post-exploitation, buffer-overflow research
Malware / Forensics	Ghidra, Volatility, GDB, Wireshark, Windows forensic artifacts, registry analysis, filesystem/disk investigation
Threat Research	MITRE ATT&CK, OSINT, threat-report analysis, TTP analysis, infrastructure analysis, adversary research
Programming / Systems	Python, Bash, C, SQL, Linux, Kali Linux, Windows, GCC, GDB, Git, Linux kernel headers, kernel build workflows

CERTIFICATIONS & PROFESSIONAL TRAINING

Google Cybersecurity Professional Certificate	Feb'26 Coursera
Security, networking, Linux, SQL, Python, SIEM workflows, threat detection, incident response, and security operations.	
Tata — Cybersecurity Analyst Job Simulation	Aug'26 Forge
IAM strategy assessment, solution design, implementation planning, and secure access workflows.	
Introduction to Blockchain and Web3	May'24 edX
Blockchain, decentralized systems, Web3 architecture, and security considerations.	
Natural Language Processing Specialization	Apr'23 Coursera
Text preprocessing, classification, and sequence-model workflows using Python.	

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Engineering	2024
Lovely Professional University	Jalandhar, Punjab
Minor: Machine Learning	Relevant study: Network Security, Ethical Hacking, Penetration Testing, Machine Learning, Automation
Intermediate (MPC), Telangana Board of Intermediate Education	2020
Mathematics, Physics, Chemistry	Warangal, Telangana

CORE SECURITY CAPABILITIES

VAPT Delivery	Reconnaissance, service enumeration, vulnerability assessment, vulnerability validation, controlled exploitation, privilege escalation, post-exploitation, attack-path analysis, technical reporting
Web Security Testing	HTTP analysis, SQL injection, XSS, authentication, authorization, access-control testing, request/response analysis, Burp Suite, OWASP ZAP
Network Analysis	Network reconnaissance, service enumeration, exposed-service analysis, packet capture, traffic analysis, protocol analysis, Nmap, Wireshark, Metasploit
Threat Research	Threat intelligence, OSINT, threat-report analysis, TTP analysis, adversary behavior, infrastructure analysis, MITRE ATT&CK
Malware / Forensics	Static/dynamic analysis, suspicious-file investigation, Windows registry artifacts, filesystem activity, disk analysis, memory investigation
Security Engineering	Linux security, wireless security, kernel compatibility, C systems programming, Python/Bash automation, Git, GCC, technical troubleshooting
Security Documentation	Technical evidence, vulnerability impact, exploitation conditions, attack paths, remediation recommendations, retesting, security reporting

AI / ML SECURITY DIRECTION

- Machine Learning minor with academic work in Natural Language Processing and a capstone project involving text classification.
- Research interest in AI security, including prompt-injection risks, RAG security, model-integrated application attack surfaces, insecure tool use, and AI-assisted security workflows.

ADDITIONAL TRAINING

- Java Full Stack Developer Training — Feb'23 | CipherSchools — Java application development and databases.
- Google Cloud Facilitator Program — May'22 | Google — cloud computing and infrastructure.

PROFESSIONAL FOCUS

Security Engineering	Cybersecurity Analysis	Security Operations	VAPT	Web Application Security
Threat Intelligence	Malware Analysis	Digital Forensics	Reverse Engineering	Security Research

Investigate the attack path. Validate the evidence. Understand the system. Improve the security.